Group 90 Progress Report:
Sleep Debt and Screen Time: Late Night Phone Habits

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1 Introduction

Smartphones are an integral part of our lives due to benefits; such as easy accessibility to information, social connectivity, workplace/financial applications, and entertainment including watching videos or playing video games. Smartphones have been revolutionized the ease of information and has everyone connected. However, the smartphone - with many pros stated - has also many drawbacks that has disconnected everyone from reality. Spending too much time on the screen reduced personal socialization and significant sleep (CBC News, 2026). Sleep plays a crucial role in developing new memory and is essential for all stages of learning. With inadequate amounts of sleep, the body and mind experience rapid physical and cognitive decline, affecting overall long-term health. These include mood swings, (Shah et al., 2024). Over 3/4 of Americans lose sleep because of digital distractions (Newswise, 2026). The number might be different for different nations, but this is a severe problem that affects a lot of people.

The goal of this project is to analyze and predict the fatigue that people feel the following day based on their screentime usage.

There are a few challenges with trying to predict 'sleep fatigue' based on screentime. There are a lot of variables involved, they might not scale linearly with sleep loss, and there is a limit to the amount of sleep fatigue a person might experience (it could be different for different people).

The data type is tabular containing number and category features like screen time, type of screen use, device type, and the resulting sleep patterns during weekdays and weekends. The task type is regression since we are trying to figure out the exact amount of sleep loss in hours. The number of classes is not applicable for this task since we are using regression. Since we are predicting sleep loss for both weekdays and weekends, this is a multi-label problem where we will get 2 outputs per set of inputs.

2 Related Work

Here, talk about the related work you encountered for your approach. Cite at least 5 references. Refer to item 2. No one has done exactly your task? Write about the most similar thing you can find. This should be around 0.25-0.5 pages.

3 Dataset and Preprocessing

3.1 Data Source and Characteristics

We're using the research dataset by Perez-Lloret and Perez-Chada (2022), hosted on the Mendeley Data repository under a Creative Commons open-access license. The survey records data collected across 52 secondary schools in Argentina. The survey captures metrics such as demographic attributes like age and gender, device access, the types of digital screentime (gaming, social media, and video streaming), sleep timing, and clinical screening tools to look for narcolepsy and other disorders.

The raw archive (`repositorio.xlsx`) contains 4,115 raw respondent submissions with 123 distinct tabular variables recorded in Spanish.

3.2 Data Cleaning and Feature Harmonization

To establish a machine learning pipeline, we first implemented a data translation pipeline:

1. **Translation and Feature Extraction:** All feature headings and the responses were mapped from Spanish to English (ex: tpopantallatotalhs -> screen_total_hrs, tporedeshs -> screen_social_media_hrs, andtposueñosem -> sleep_weekday_hrs).

2. **Cohort Filtering and Missingness:** We isolated the participants from 11 to 20 years.

Records with missing values for primary sleep durations or total screen hours were eliminated. Sub-activity values (such as gaming or console hours) where respondents indicated no use were set to 0 hours.

3. **Clinical Plausibility Bounding:** Outlier responses that did not make physical sense for humans were excluded (restricting sleep durations to $3.0 \leq$ sleep ≤ 16.0 hours/day and daily screen exposure to $0.0 \leq$ screen ≤ 24.0 hours/day).

After this preprocessing, our finalized dataset is composed of 2,496 student records, which have been exported to /dataset_clean_english.csv.

3.3 Target Formulation: Sleep Debt

Consistent with pediatrician recommendations made by Paruthi et al. (2016), the target sleep duration for adolescents is set to $S_{\text{rec}} = 9.0$ hours. We define the target vector $\mathbf{y} = [y_{\text{weekday}}; y_{\text{weekend}}] \in \mathbb{R}^2$ as:

$$y_{\text{weekday}} = 9.0 - T_{\text{sleep, weekday}} \tag{1}$$

$$y_{\text{weekend}} = 9.0 - T_{\text{sleep, weekend}} \tag{2}$$

Under this formula, positive values represent severe sleep loss (sleep debt), whereas negative values represent sleep surplus. In our preprocessed data, adolescents experience an average weekday sleep loss of $\mu = 1.96$ hours ($\sigma = 1.36$ hours), contrasted with a weekend sleep surplus of $\mu = -0.42$ hours ($\sigma = 1.91$ hours), with a mean social jetlag of 2.38 hours of rebound sleep on weekends.

4 Features

Describe any features you used for your model, or how your data was input to your model. Are you doing feature engineering or feature selection? Are you learning embeddings? Is it all part of one neural network? Refer to item 2. This may range from 0.25 pages to 0.5 pages.

5 Model Implementation

We represent our task as a multi-target linear and non-linear regression problem where we learn a parameter mapping $f : \mathbb{R}^{13} \to \mathbb{R}^2$ optimizing the Mean Squared Error (MSE):

$$MSE(W) = \frac{1}{N} \sum_{i=1}^{N} \|\mathbf{y}_i - \hat{\mathbf{y}}_i\|_2^2 \tag{3}$$

To establish benchmarks and evaluate model learning, we implemented three candidate solutions with scikit-learn:

5.1 Baseline 1: Empirical Mean Regressor

We construct a naive baseline predicting the training set mean vector for every test instance:

$$\hat{\mathbf{y}}_{\text{dummy}} = \bar{\mathbf{y}}_{\text{train}} = \frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} \mathbf{y}_i \tag{4}$$

This baseline represents the lower bound ($R^2 \approx 0.0$).

5.2 Baseline 2: Multiple Linear Regression (OLS)

We use a multi-output Ordinary Least Squares (OLS) solution:

$$\hat{\mathbf{y}} = \mathbf{W}^T \tilde{\mathbf{x}} + \mathbf{b} \tag{5}$$

5.3 Candidate Model: Polynomial Regression with Ridge Regularization

Recognizing that screen exposure exhibits nonlinear effects, we construct a degree-2 polynomial feature mapping $\phi(\mathbf{x})$ encompassing all linear, quadratic, and pairwise interaction terms:

$$\phi(\mathbf{x}) = [1, x_1, \ldots, x_d, x_1^2, x_1 x_2, \ldots, x_d^2]^T \in \mathbb{R}^{104} \tag{6}$$

To prevent overfitting on the 104-dimensional space, we use the Ridge objective with penalty parameter $\alpha = 100.0$:

$$L_{\text{Ridge}} = \text{MSE} + \alpha \|\mathbf{w}\|_2^2 \tag{7}$$

6 Results and Evaluation

6.1 Evaluation Protocol

We split the preprocessed dataset N = 2,496 using an 80/20 train-test split with fixed random seed ($N_{\text{train}} = 1{,}996$, $N_{\text{test}} = 500$). Performance is measured across three standard regression metrics:

$$\text{MAE} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} |y_i - \hat{y}_i| \tag{8}$$

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (y_i - \hat{y}_i)^2} \tag{9}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \tag{10}$$

Model Architecture	Weekday Sleep Loss (Hours)			Weekend Sleep Loss (Hours)		
	MAE ↓	RMSE ↓	R^2 ↑	MAE ↓	RMSE ↓	R^2 ↑
Baseline 1: Mean Regressor	1.043	1.406	-0.000	1.549	1.949	-0.002
Baseline 2: Linear Regression (OLS)	**0.971**	**1.332**	**0.103**	**1.511**	**1.920**	**0.027**
Candidate: Polynomial Ridge (Deg 2, $\alpha = 100$)	0.988	1.360	0.065	1.535	1.967	-0.021

Table 1: Preliminary quantitative performance across 500 held-out test adolescents. Metrics report Mean Absolute Error (MAE in hours), Root Mean Squared Error (RMSE in hours), and Coefficient of Determination (R^2).

6.2 Preliminary Findings

The test results are detailed in Table 1. Multiple Linear Regression achieves a Mean Absolute Error of 0.971 hours on weekday sleep loss ($\hat{R2}$ = 0.103, RMSE = 1.332 hours), demonstrating that it's better than the baseline (MAE $= 1.043$ hours).

The degree-2 polynomial model with Ridge regularization ($\alpha = 100$) has a slightly higher test MAE (0.988 hours) than standard linear regression. This suggests that expanding to 104 quadratic and interaction features adds noise that outweighs non-linear gains when constrained to a linear regularizer.

Furthermore, all models showed lower predictive accuracy on weekend sleep loss ($R^2 \approx 0.027$) compared to weekday sleep loss ($R^2 \approx 0.103$). Weekend sleep schedules among teenagers show weirder behaviour and compensatory rebound sleeping, which is weakly correlated with isolated screen time.

7 Feedback and Plans

Write about your plans for the remainder of the project. This should include a discussion of the feedback you received from your TA, and how you plan to improve your approach. Reflect on your implementation and areas for improvement. Refer to item 6. This may be around 0.5 pages.

Team Contributions

Write in this section a few sentences describing the contributions of each team member. What did each member work on? Refer to item 7.

References

CBC News. 2026. Kids and young adults are addicted to screens. New research argues it may be a public health concern. https://www.cbc.ca/news/health/youth-screen-time-tech-companies-9.7365434.

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Shalini Paruthi, Lee J. Brooks, Carolyn D'Ambrosio, Wendy A. Hall, Suresh Kotagal, Robin M. Lloyd, Beth A. Malow, Kiran Maski, Cynthia Nichols, Stuart F. Quan, and 1 others. 2016. Recommended amount of sleep for pediatric populations: A consensus statement of the American Academy of Sleep Medicine. *Journal of Clinical Sleep Medicine*, 12(6):785–786.

Santiago Perez-Lloret and Daniel Perez-Chada. 2022. Screen use, sleep duration, daytime somnolence, and academic failure in school-aged adolescents. *Mendeley Data, V1*.

A. S. Shah and 1 others. 2024. Effects of sleep deprivation on physical and mental health outcomes: An umbrella review. *American Journal of Lifestyle Medicine*.